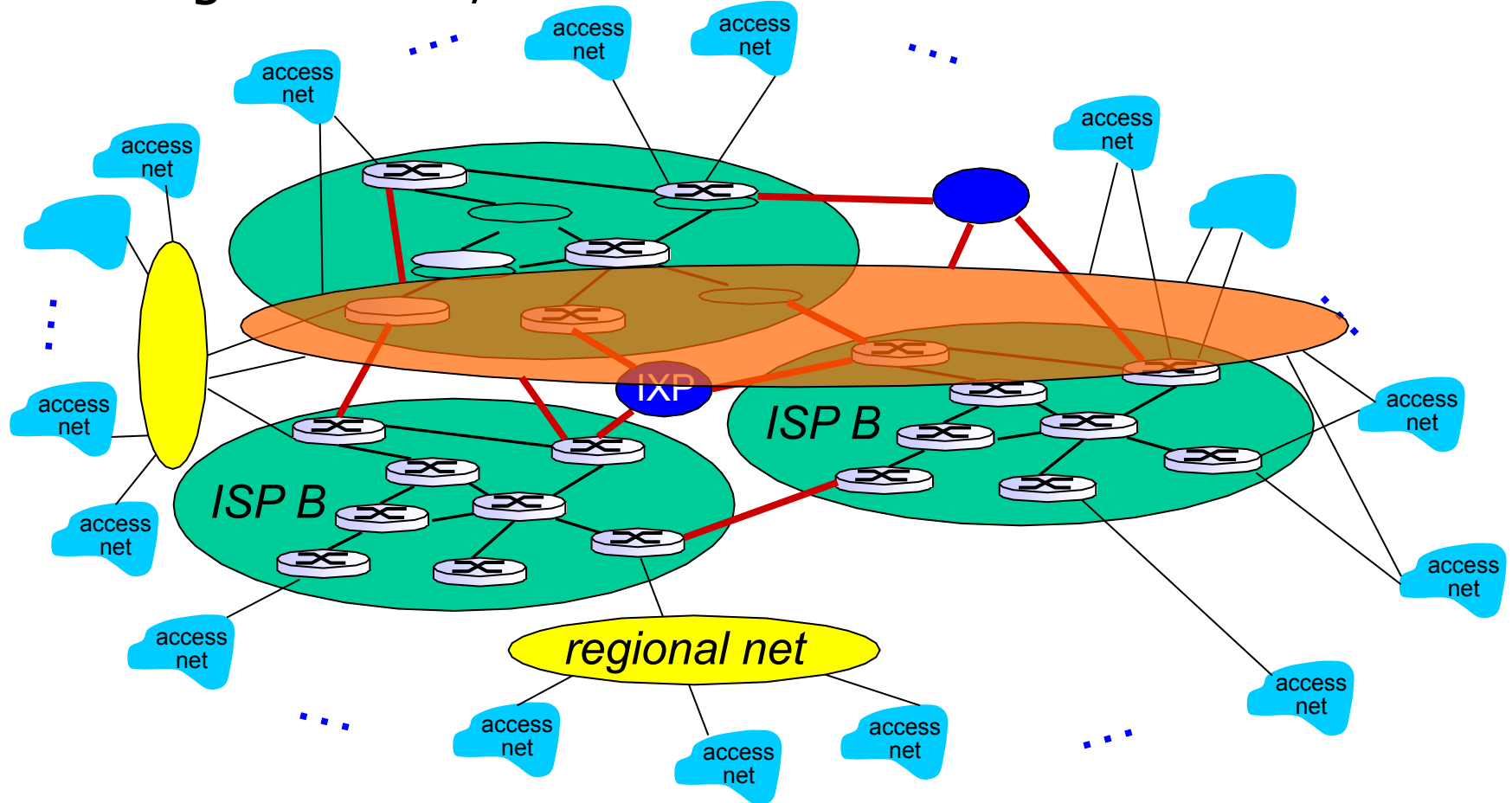
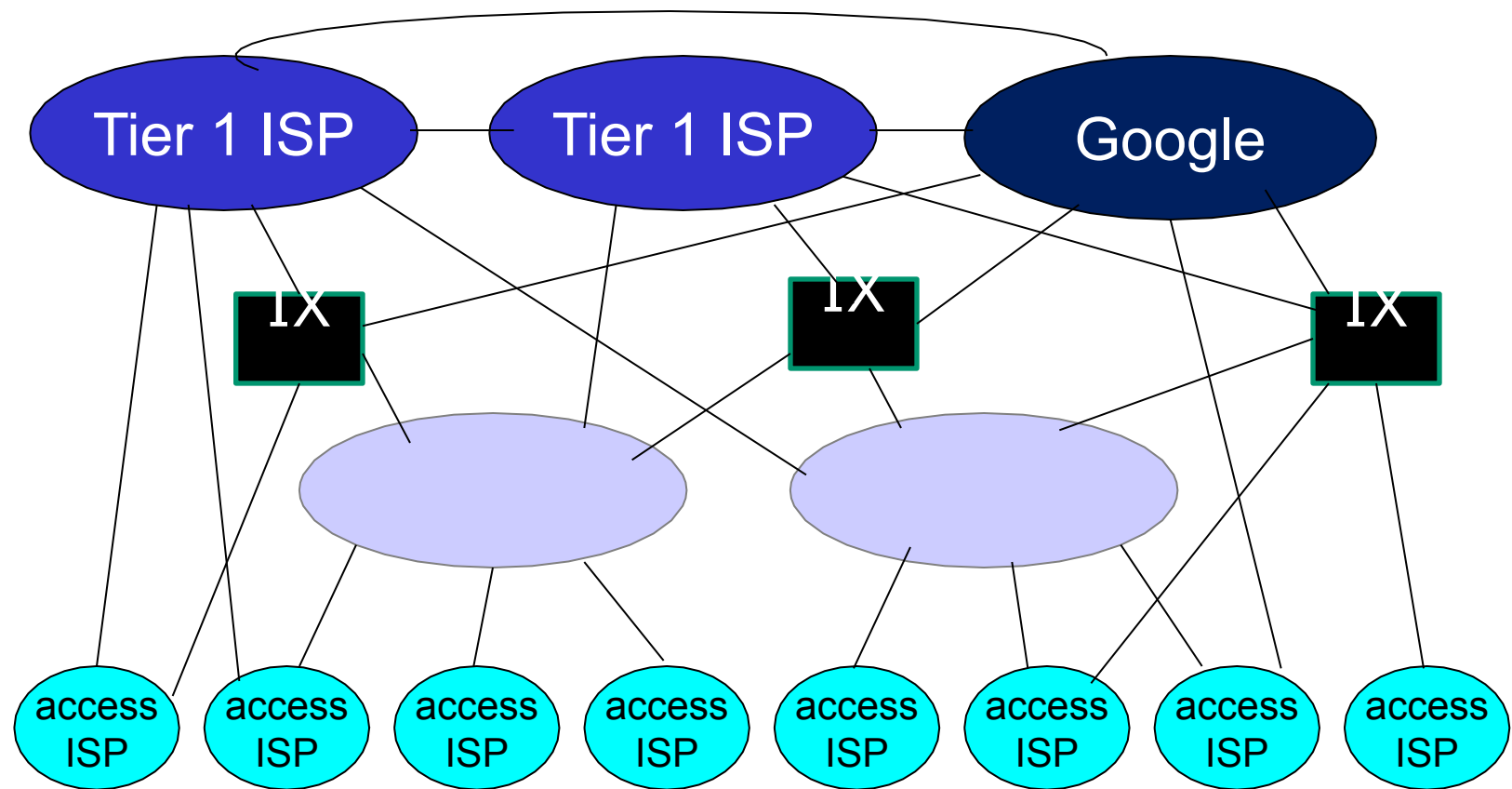


Internet structure: network of networks

... and content provider networks (e.g., Google, Microsoft, Akamai) may run their own network, to bring services, content close to end users



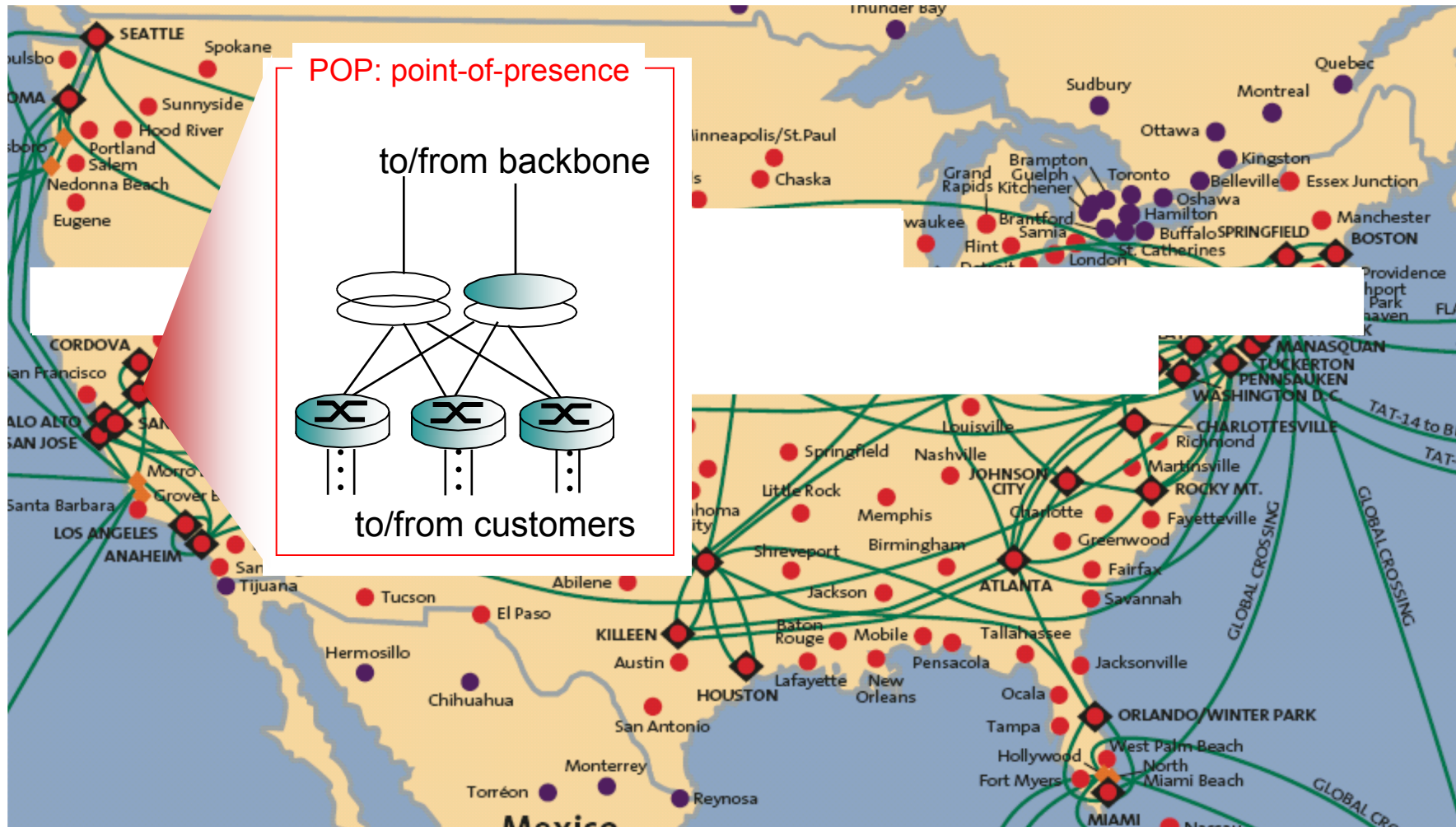
Internet structure: network of networks



❖ at center: small # of well-connected large networks

- **"tier-1" commercial ISPs** (e.g., Level 3, Sprint, AT&T, NTT), national & international coverage
- **content provider network** (e.g., Google): private network that connects its data centers to Internet, often bypassing

Tier-1 ISP: e.g., Sprint



Chapter 1: roadmap

1.1 what *is* the Internet?

1.2 network edge

- end system

networks, links

-

network

structure

1.4 delay, loss, throughput in networks

1.5 protocol layers, service models

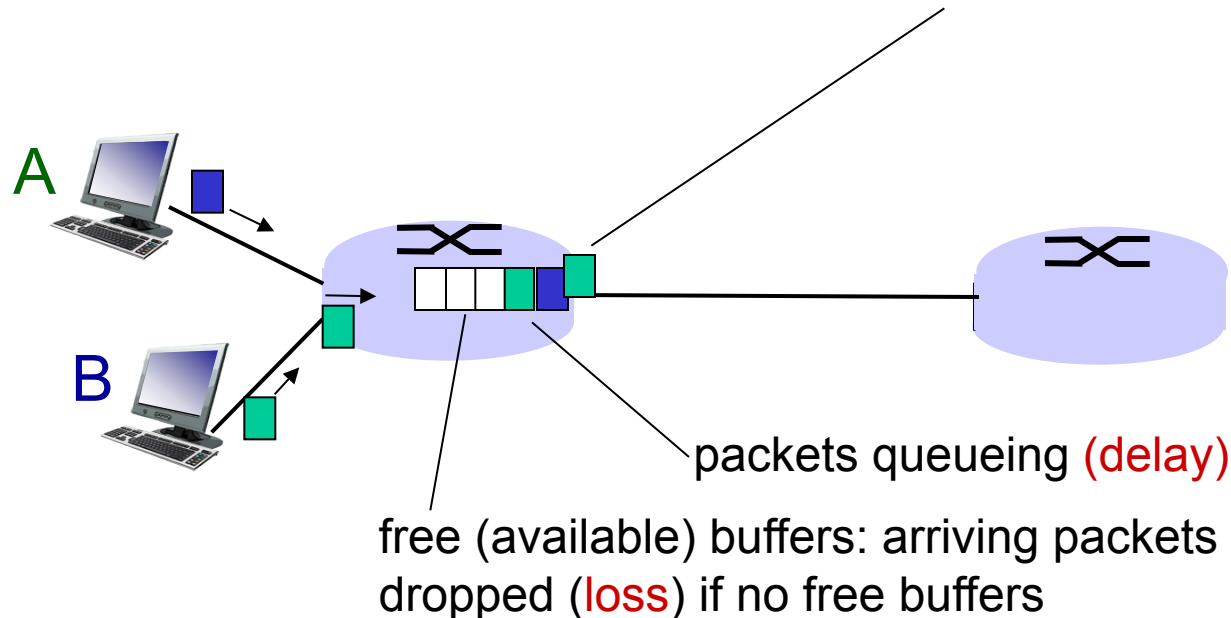
1.6 networks under attack: security

1.7 history

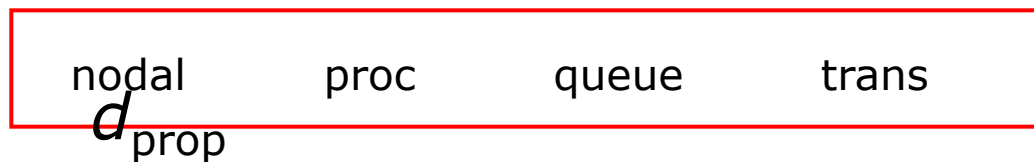
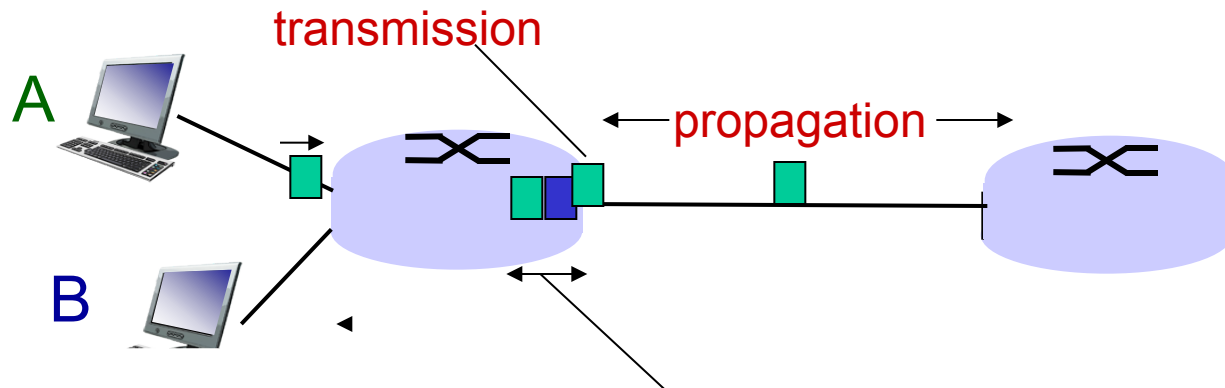
How do loss and delay occur?

packets *queue* in router buffers

- ❖ packet arrival rate to link (temporarily) exceeds output link capacity
- ❖ packets queue,



Four sources of packet delay



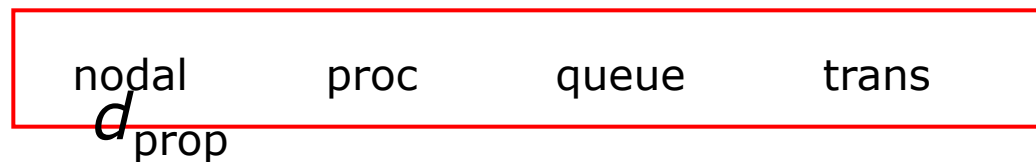
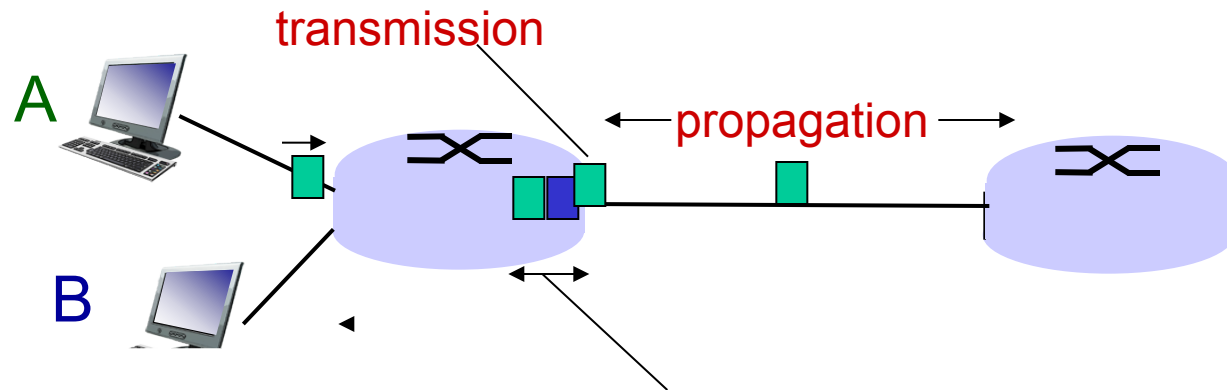
d_{proc} : nodal processing

- check bit errors
- determine output link
- typically < msec

d_{queue} : queueing delay

- time waiting at output link for transmission
- depends on congestion level of

Four sources of packet delay



d_{trans} : transmission delay:

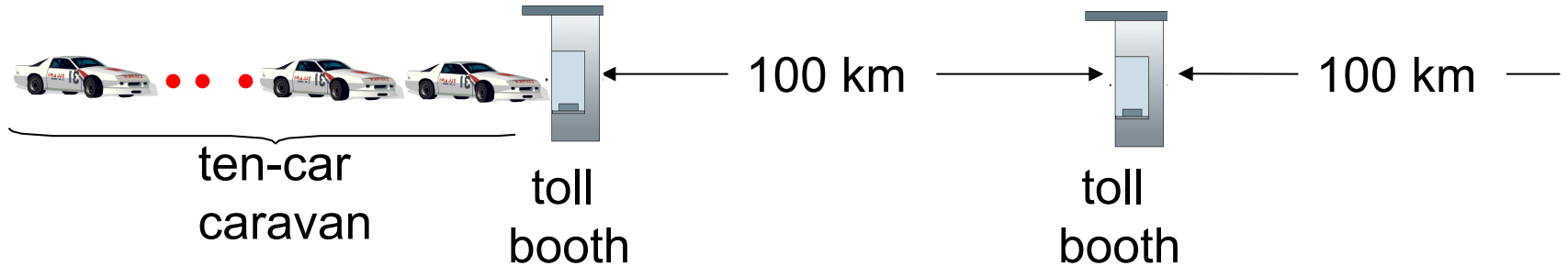
- L : packet length (bits)
 - R : link bandwidth (bps)
 - $d_{trans} = L/R$
- d_{trans} and d_{prop} very different

d_{prop} : propagation delay:

- d : length of physical link
- s : propagation speed in medium ($\sim 2 \times 10^8$ m/sec)
- $d_{prop} = d/s$

* Check out the Java applet for an interactive animation on trans vs. prop delay

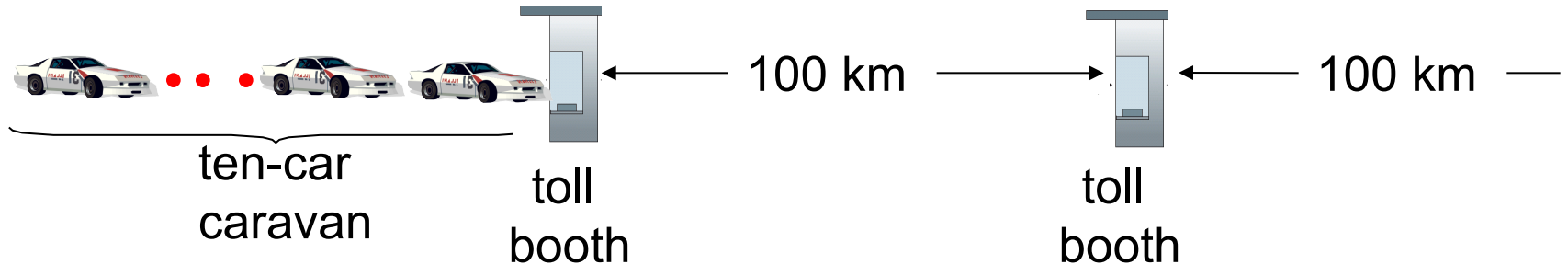
Caravan analogy



- ❖ toll booth to service car (bit transmission time)
- ❖ car $\sim$ bit; caravan $\sim$ packet
- ❖ **Q: How long until caravan is lined up before 2nd toll booth?**

- II booth onto highway = $12 \times 10 = 120$ sec
- time for last car to propagate from 1st to 2nd toll booth:
 $100\text{km} / (100\text{km/hr}) = 1$ hr
- **A: 62 minutes**

Caravan analogy (more)



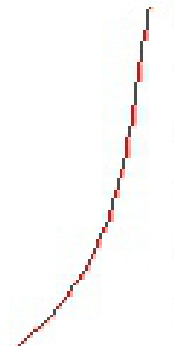
servic

- ❖ Q: Will cars arrive to 2nd booth before all cars serviced at first booth?
 - A: Yes! after 7 min, 1st car arrives at second booth; three cars still at 1st booth.

Queueing delay (revisited)

- ❖ R : link bandwidth (bps)
- ❖ L : packet length

queueing
delay



→ La/R

arrival

$= La/R$

- ❖ $La/R \sim 0$: avg. queueing delay small
- ❖ $La/R \rightarrow 1$: avg. queueing delay large
- ❖ $La/R > 1$: more "work" arriving than can be serviced, average delay infinite!



$La/R \sim 0$



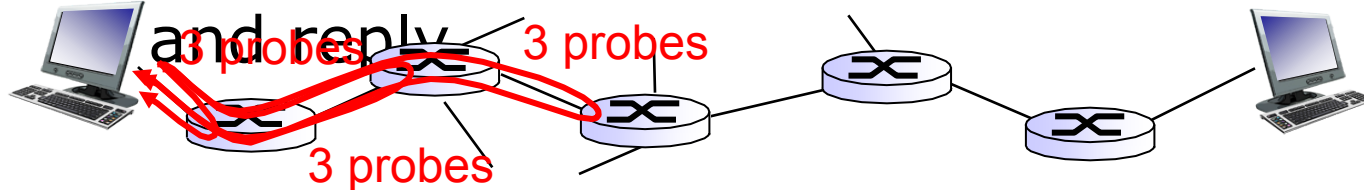
$La/R \rightarrow 1$

* Check out the Java applet for an interactive animation on queueing and loss

“Real” Internet delays and routes

- ❖ what do “real” Internet delay & loss look like?
- ❖ `traceroute` program: provides delay measurements to router

- send probe packets to router i on path towards destination
- router i will return packets to sender
- sender times interval between transmission and reply



"Real" Internet delays, routes

traceroute: gaia.cs.umass.edu to www.eurecom.fr

3 delay measurements from
gaia.cs.umass.edu to cs-gw.cs.umass.edu



```
1 cs-gw (128.119.240.254) 1 ms 1 ms 2 ms
2 border1-rt-fa5-1-0.gw.uma s 1 ms 2 ms
3 cht-vbns.gw.umass.edu (1 ms

6 abilene-v
7 nycm-wa
8 62.40.103.253 (62.40.103.253) 104 ms 109 ms 106 ms
9 de2-1.de1.de.geant.net (62.40.96.129) 109 ms 102 ms 104 ms
10 de.fr1.fr.geant.net (62.40.96.50) 113 ms 121 ms 114 ms
11 renater-gw.fr1.fr.geant.net (62.40.103.54) 112 ms 114 ms 112 ms
12 nio-n2.cssi.renater.fr (193.51.206.13) 111 ms 114 ms 116 ms
13 nice.cssi.renater.fr (195.220.98.102) 123 ms 125 ms 124 ms
14 r3t2-nice.cssi.renater.fr (195.220.98.110) 126 ms 126 ms 124 ms
15 eurecom-valbonne.r3t2.ft.net (193.48.50.54) 135 ms 128 ms 133 ms
16 194.214.211.25 (194.214.211.25) 126 ms 128 ms 126 ms
17 * * *
18 * * *
19 fantasia.eurecom.fr (193.55.113.142) 132 ms 128 ms 136 ms
```

rans-oceanic link

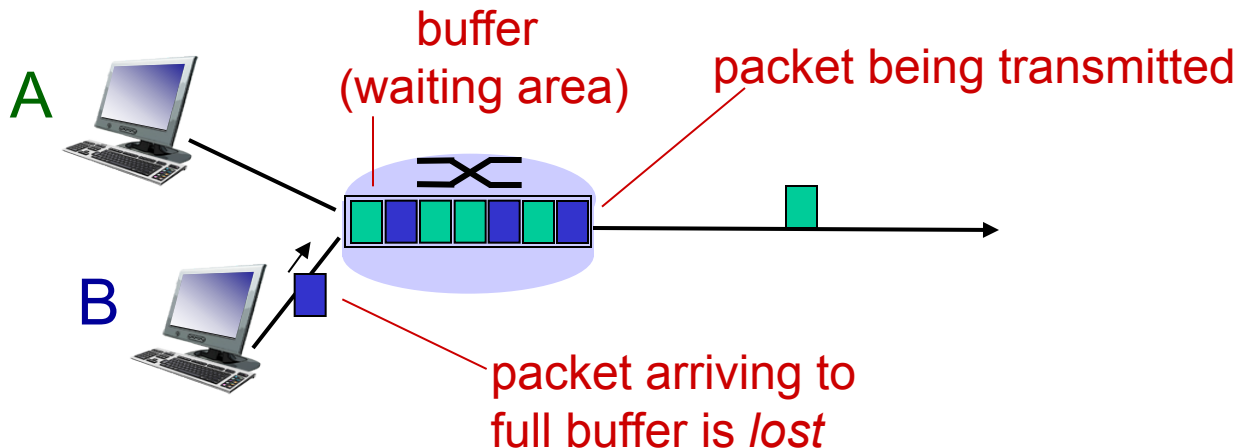
* means no response (probe lost, router not replying)

* Do some traceroutes from exotic countries at www.traceroute.org

Packet loss

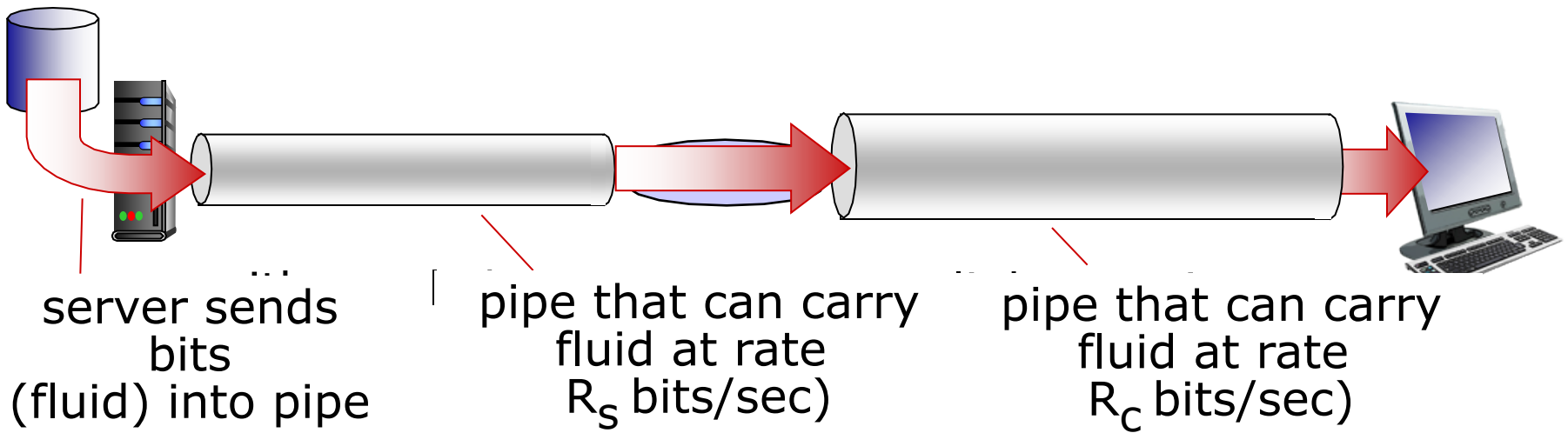
- ❖ queue (aka buffer) preceding link in buffer has finite capacity
- ❖ packet arriving to full queue dropped (aka lost)
- ❖

not at all



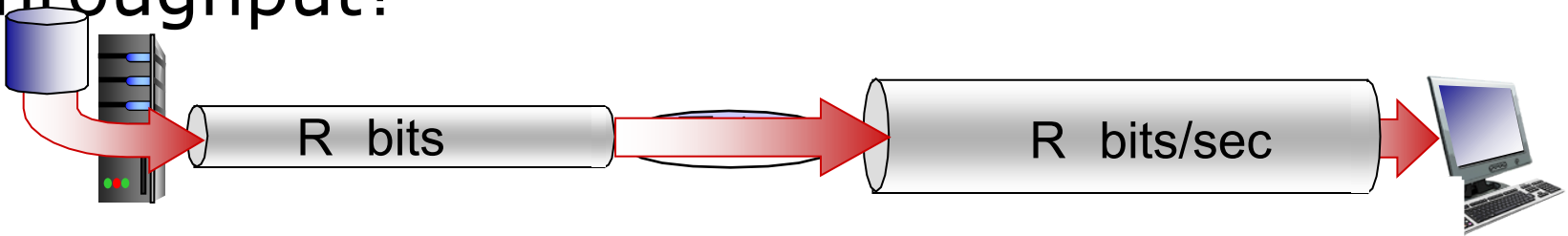
Throughput

- ❖ *throughput*: rate (bits/time unit) at which bits transferred between sender/receiver
 - *instantaneous* point in time

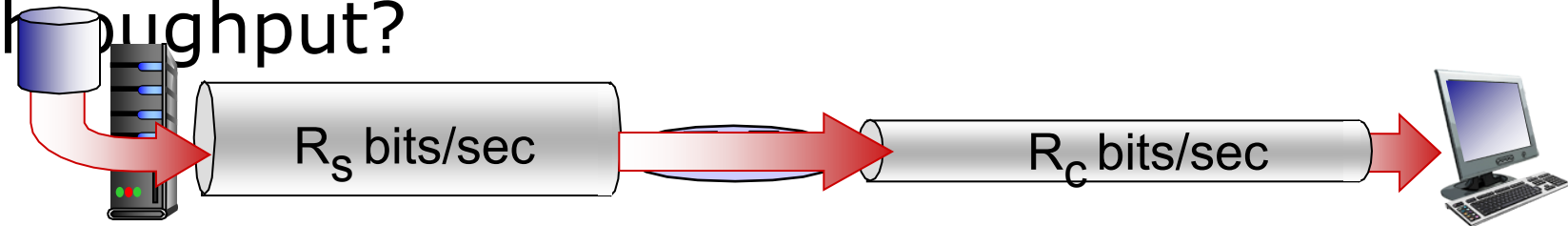


Throughput (more)

- ❖ $R_s < R_c$ What is average end-end throughput?



- ❖ $R_s >$
throughput?



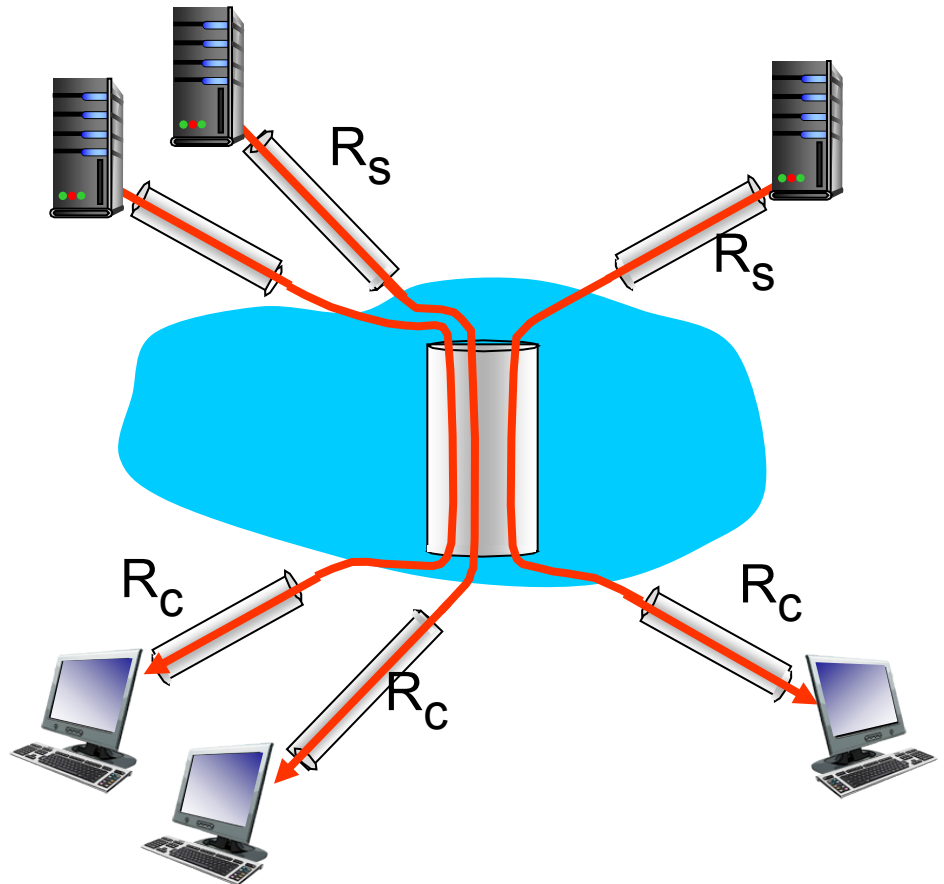
bottleneck

link on end-end path that constrains end-end throughput

Throughput: Internet scenario

- ❖ per-connection end-end throughput:

R_s is bottleneck



10 connections (fairly) share backbone bottleneck link R bits/sec

Chapter 1: roadmap

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1.2 network edge

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1.7 history

Protocol “layers”

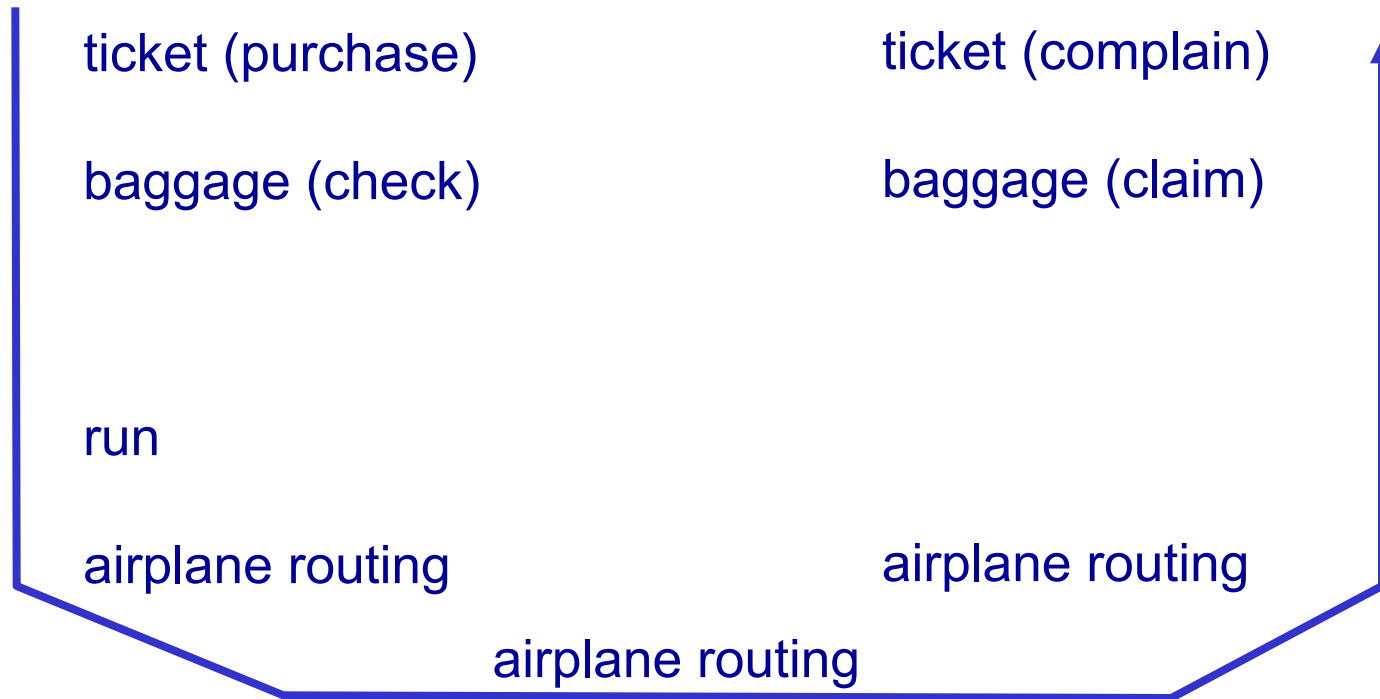
*Networks are
complex,
with many
“pieces”:*

- links of various media
- applications
- protocols
- hardware, software

Question:

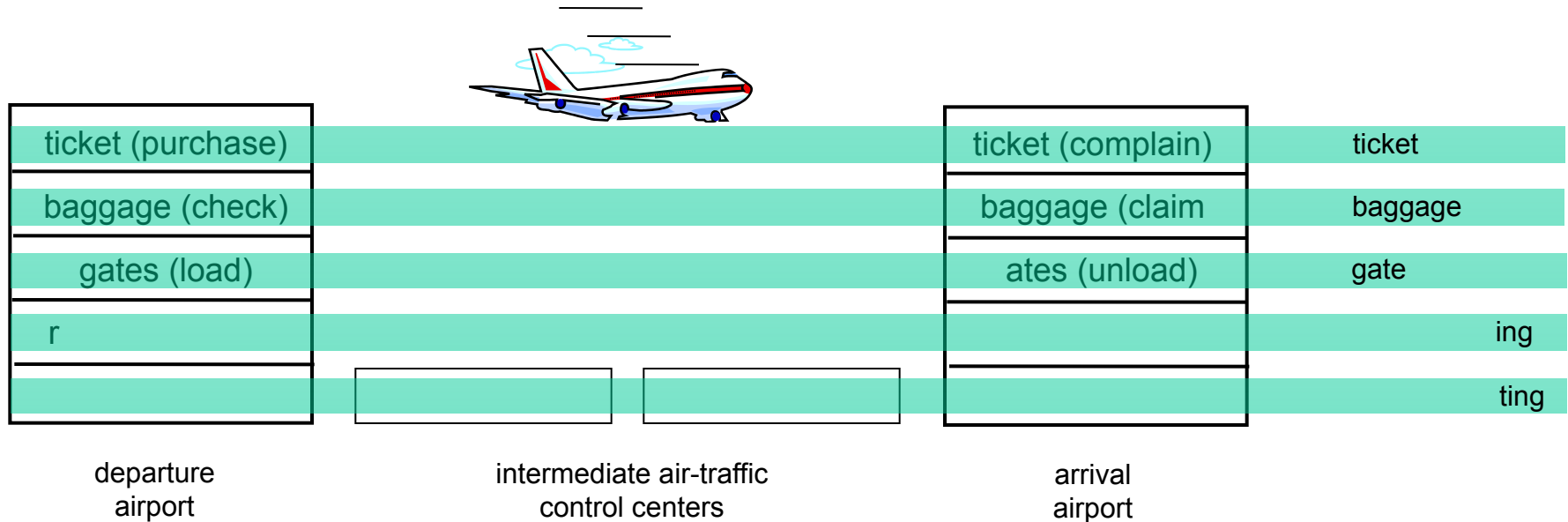
of
network?

.... or at least our
discussion of
networks?



❖ a series of steps

Layering of airline functionality



layers: each layer implements a service

- via its own internal-layer actions
- relying on services provided by layer below

Why layering?

dealing with complex systems:

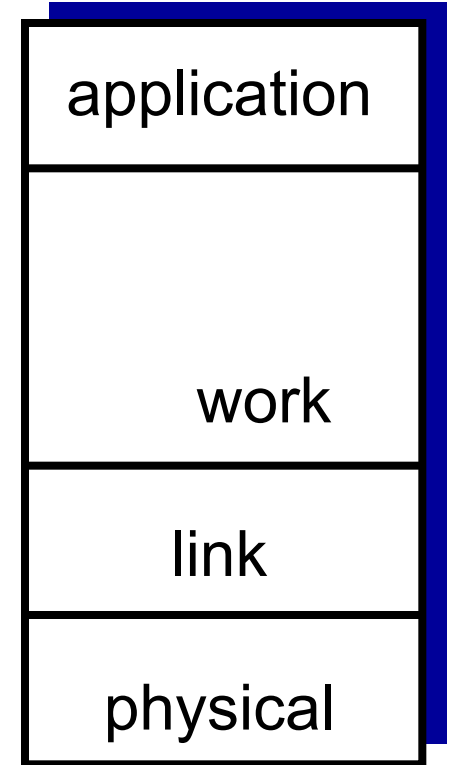
- ❖ explicit structure allows identification, relationship of components and their pieces

updates

- change of implementation of layer's service transparent to rest of system
- e.g., change in gate procedure doesn't affect rest of system
- ❖ layering considered harmful?

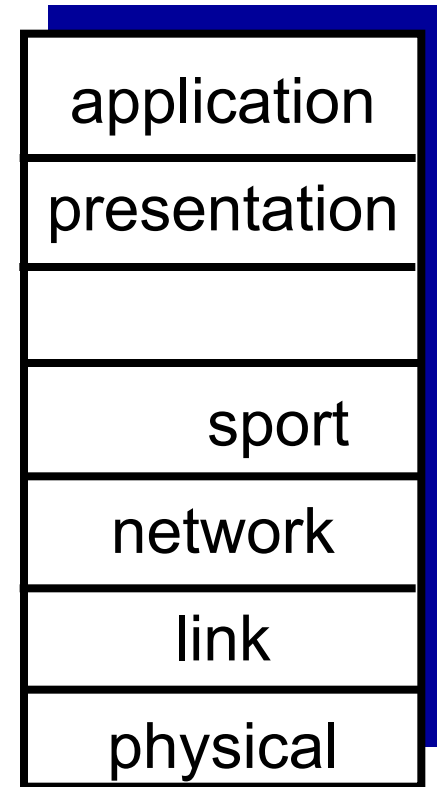
Internet protocol stack

- ❖ *application*: supporting network applications
 - FTP, SMTP, HTTP
- ❖ *transport*: pr
- ❖ *netw*
datagrams from source to destination
 - IP, routing protocols
- ❖ *link*: data transfer between neighboring network elements
 - Ethernet, 802.111 (WiFi), PPP
- ❖ *physical*: bits “on the wire”

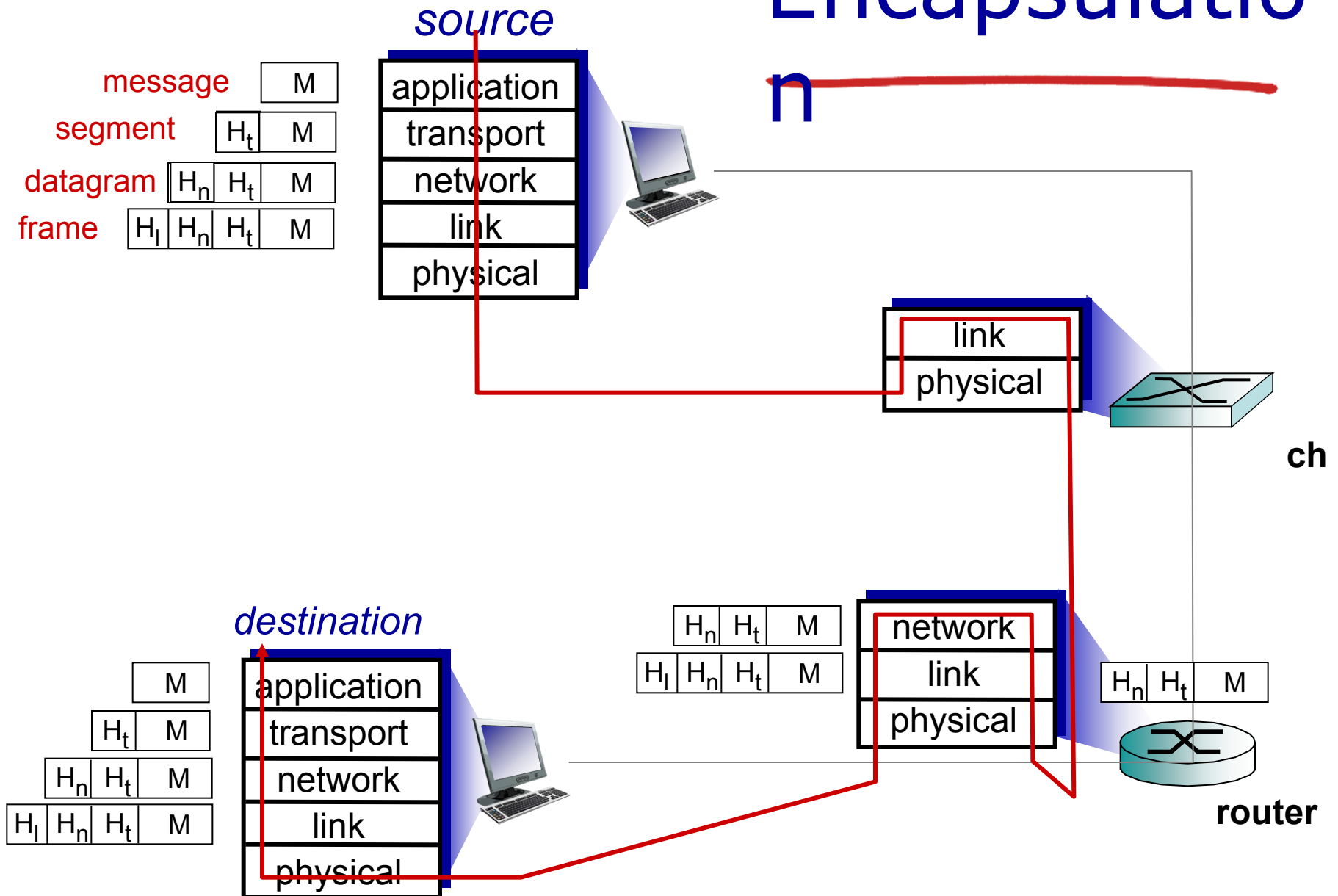


ISO/OSI reference model

- ❖ *presentation*: allow applications to interpret meaning of data, e.g., encryption, c
- ❖ *session*: synchronization, checkpointing, recovery of data exchange
- ❖ Internet stack “missing” these layers!
 - these services, *if needed*, must be implemented in application
 - needed?



Encapsulation



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Network security

❖ field of network security:

- how bad guys can attack computer networks
- how we can defend against

im

- ❖ Internet not originally designed with (much) security in mind

- *original vision*: “a group of mutually trusting users attached to a transparent network” 😊
- Internet protocol designers playing “catch-up”
- security considerations in all layers!

Bad guys: put malware into hosts via Internet

- ❖ malware can get in host from:
 - *virus*: self-replicating infection by receiving/executing object (e.g., e-mail attachment)
- rec ed
- ❖ *spyware malware* can record keystrokes, web sites visited, upload info to collection site
- ❖ infected host can be enrolled in *botnet*, used for spam. DDoS attacks

Bad guys: attack server, network infrastructure

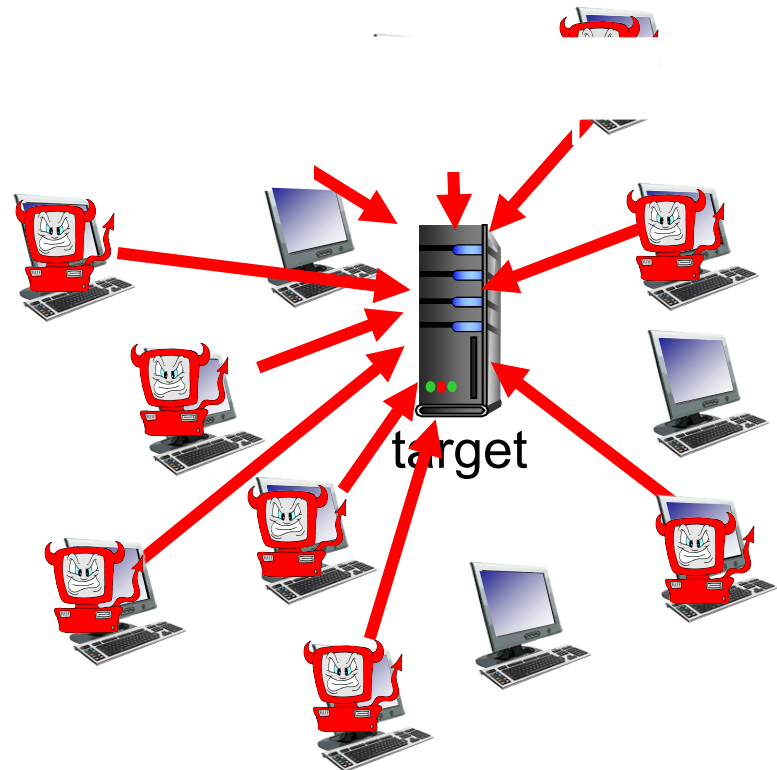
Denial of Service (DoS): attackers make resources (server, bandwidth) unavailable to legitimate traffic by overwhelming resource with

1

2

around

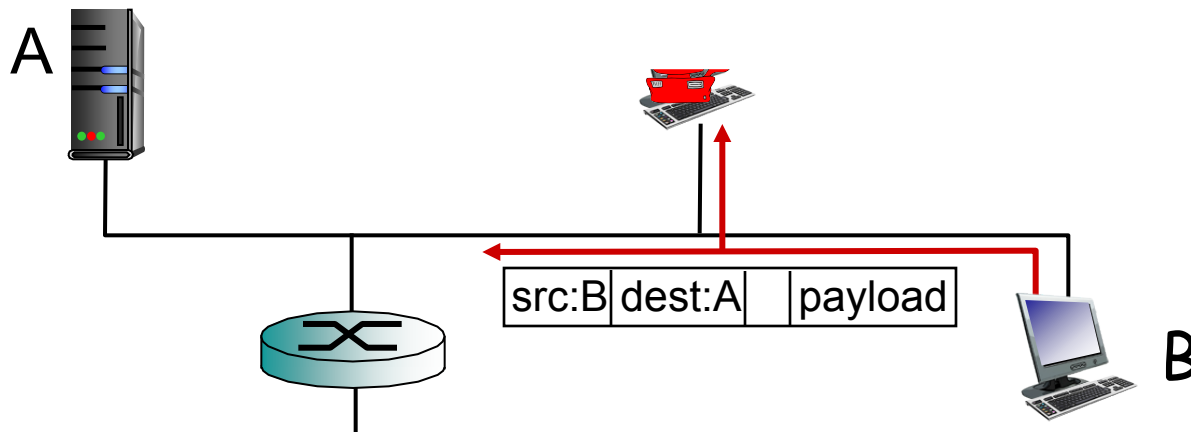
3. (see botnet)
send packets to target
from compromised
hosts



Bad guys can sniff packets

packet "sniffing":

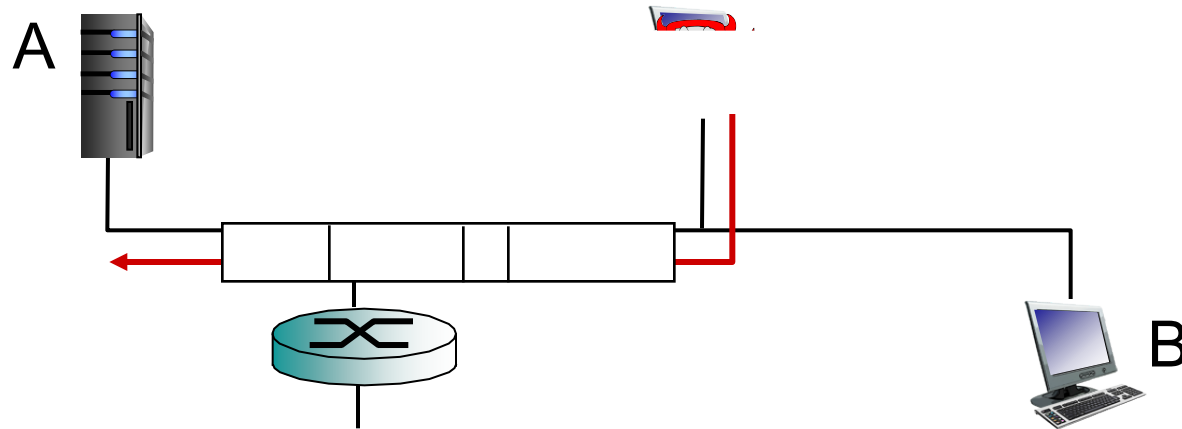
- broadcast media (shared ethernet, wireless)
- promiscuous network interface reads/records all packets (e.g., passwords!)



- ❖ wireshark software used for end-of-chapter labs is a (free) packet-sniffer

Bad guys can use fake addresses

IP spoofing: send packet with false source address



... lots more on security (throughout, Chapter

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Internet history

1961-1972: Early packet-switching principles

❖ 1961: Kleinrock - queueing theory shows effectiveness

❖ 1972:

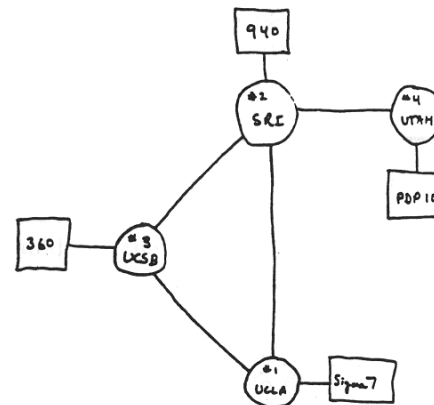
- ARPAnet public demo (Network Control Program)

packet
military nets

❖ 1967: ARPAnet conceived by Advanced Research Projects Agency

❖ 1969: first ARPAnet node operational

- ARPAnet has 15 nodes



THE ARPA NETWORK

Internet history

1972-1980: Internetworking, new and proprietary nets

- ❖ 1970: ALOHAnet satellite network in Hawaii
- ❖ 1974: Cerf and K
- ❖ 1976: PARC
- ❖ late70's: proprietary architectures: DECnet, SNA, XNA
- ❖ late 70's: switching fixed length packets (ATM precursor)
- ❖ 1979: ARPAnet has 200 nodes

Cerf and Kahn's internetworking

changes
required to
interconnect networks

- best effort service model
- stateless routers
- decentralized control

define today's Internet
architecture

Internet history

1980-1990: new protocols, a proliferation of networks

- ❖ 1983: deployment of TCP/IP
- ❖ 1982: smtp e-mail
- ❖ 1985: ftp protocol defined
- ❖ 1988: TCP congestion control
- ❖ new national networks: Cernet, NSFnet, for national address translation
- ❖ confederation of networks

Internet history

1990, 2000's: commercialization, the Web, new apps

❖ early 1990's: ARPAnet decommissioned

❖ 1991: NSF lifts

late 1990's – 2000's:

❖ more killer apps: instant messaging,

(decom

❖ early 1990s: Web

- hypertext [Bush 1945, Nelson 1960's]

- HTML, HTTP: Berners-Lee

- 1994: Mosaic, later Netscape

- late 1990's: commercialization of the

❖ est. 50 million hosts, 100 million+ users

❖ backbone links running at Gbps

Internet history

2005-present

- ❖ ~750 million hosts
 - Smartphones and tablets
- ❖ Aggressive deployment of broadband access
- ❖ Emergence of new services
 - Facebook: soon one billion users
- ❖ Service providers (Google, Microsoft) create their own networks
 - Bypass Internet, providing “instantaneous” access to search, email, etc.
- ❖ E-commerce, universities, enterprises running their services in “cloud” (eg, Amazon EC2)

Introduction: summary

covered a "ton" of material!

- ❖ Internet overview
- ❖
 - pac versus circuit-switching
 - Internet structure
- ❖ performance: loss, delay, throughput
- ❖ layering, service models
- ❖ security
- ❖ history

you now have:

- ❖ context, overview, "feel" of networking
- ore depth, detail